Objectives

- ✓ Control Statements in Java.

Iteration Statements

Java's iteration statements are for, while, and do-while.

while

- ✓ The **while** loop is Java's most fundamental loop statement.
- ✓ It repeats a statement or block while its controlling expression is true.
- ✓ Here is its general form:

```
while(condition){
// body of loop
}
```

An Example

```
class While
{
    public static void main(String args[])
    {
        int n = 10;
        while(n > 0)
        {
            System.out.println("tick " + n);
            n--;
        }
    }
}
```

Output

```
tick 10
tick 9
tick 8
tick 7
tick 6
tick 5
tick 4
tick 3
tick 2
tick 1
```

do-while

- ✓ The **do-while** loop always executes its body at least once, because its conditional expression is at the bottom of the loop.
- ✓ **General Form**

```
do {
// body of loop
} while (condition);
```

An Example

```
//Demonstrate the do-while loop.
class DoWhile {
public static void main(String args[]) {
    int n = 10;
    do {
            System.out.println("tick " + n);
            n--;
        } while(n > 0);
    }
}
```

for loop

General form of the traditional **for** statement:

```
for(initialization; condition; iteration)
{
        // body
}
```

for loop

```
// Demonstrate the for loop.
class ForTick {
        public static void main()
        {
                int n;
                for(n=10; n>0; n--)
                        System.out.println("tick " + n);
        }
}
```

for each loop

- ✓ A for-each style loop is designed to cycle through a collection of objects, such as an array, in strictly sequential fashion, from start to finish.
- ✓ **General form** of the for-each loop

```
for(type itr-var : collection)
 statement-block
```

for each loop

```
int nums[] = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 };

int sum = 0;

for(int x: nums)
        sum += x;
```

Takeaways

- ✓ while loop.
- ✓ do-while loop.
- ✓ for loop.
- ✓ for each loop.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018*